SYSTEM OVERVIEW

ASTRELIS NAV (SWARM PCB) is the electronics stack for an autonomous underwater vehicle (AUV) intended to operate as part of a swarm. It senses its own orientation, motion, and depth (Sensor board), processes this data and runs navigation/control logic (Control board), communicates with other units or a base station acoustically via an underwater piezo transducer (TX/RX board), drives its motors/servos/lights for propulsion and steering (Output board), and is powered from an onboard battery with regulated multi-rail power distribution (Power board) — all built as a compact stacked-board assembly suited to a small underwater housing.

Splitting SWARM into five functional boards (Control, Power, Sensor, TX/RX, Output) stacked via headers — rather than one monolithic board or fully separate, individually-housed boards — balances modularity with integration:

Noise isolation — Power (switching regulators) and TX/RX (sensitive RX analog, H-bridge) are separated from Sensor/Control, with isolated GND domains joined only at the stack connectors, limiting noise coupling.

Compact footprint — Stacking keeps the assembly small and tube-friendly, ideal for a compact vehicle.

Simple interconnect — Shared 47-pin headers pass signals board-to-board, avoiding custom wiring harnesses.

Modularity — Any board can be revised, repaired, or swapped independently without redesigning the whole system.

Easier fab/test — Each board can be built and tested on its own before final stack assembly.

CONTROL

CONTROL — System brain

STM32F446RET6 + Raspberry Pi Pico 2 run main flight logic and higher-level processing. Includes USB-C input (with ESD protection), microSD logging, SWD programming header, RGB/status LEDs, 16MHz/32.768kHz crystals, and a boot-mode jumper for the Pico.

POWER

POWER — Energy management

Takes battery input (XT30, 4S2P pack) with surge/reverse protection, regulates to 3.3V, 5V, and 12V via buck converters, plus a boosted VPiezo rail (TPS55340) for the transducer. Includes voltage-divider power monitoring, status LEDs, and supplies power-good signals to the rest of the stack.

SENSOR

SENSOR — Motion & environmental sensing

I2C cluster: two IMUs (ASM330LHHXTR, LSM6DSO32TR), a magnetometer(MMC5603NJ), and a depth/pressure sensor (MS583730BA01-50). Shares one I2C bus with interrupt lines back to Control, which it feeds with orientation/environmental data.

TX_RX

TX/RX — Acoustic communication

Two half-bridge MOSFET drivers (UCC27211) form an H-bridge driving the piezo transducer to transmit, with a precharge sequence at power-up. On receive, a preamp/instrumentation amp (INA823) and bandpass filter (20-30kHz, for FSK) feed a comparator; a T/R switch (ADG7421) isolates TX/RX paths, and an INA240 senses H-bridge current.

OUTPUT

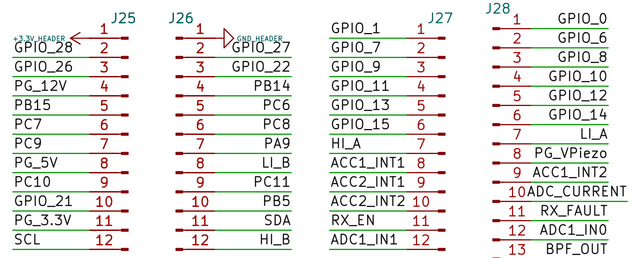
OUTPUT — Actuation

Breaks out GPIO to ESC, servo, and light headers (5V rail, per-channel pull resistors), routing Control's commands to motors/actuators to actually move the vehicle.

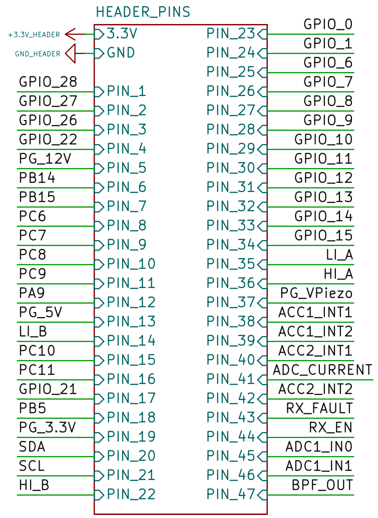
HEADER_PINS — Test/Probe Access

Breaks out stacking connector signals (J25-J28) to accessible pins for bench testing, mirroring each board's header pinout (3.3V, GND, GPIO, comms lines) without altering the signal path — lets you probe inter-board signals without disassembling the stack.

BOARD BREAKOUTS
for testing

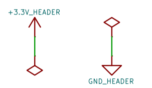


HEADER PINS
stacking connections between boards



DUPLICATE FOOTPRINTS BETWEEN BOARDS

PWR FLAGS

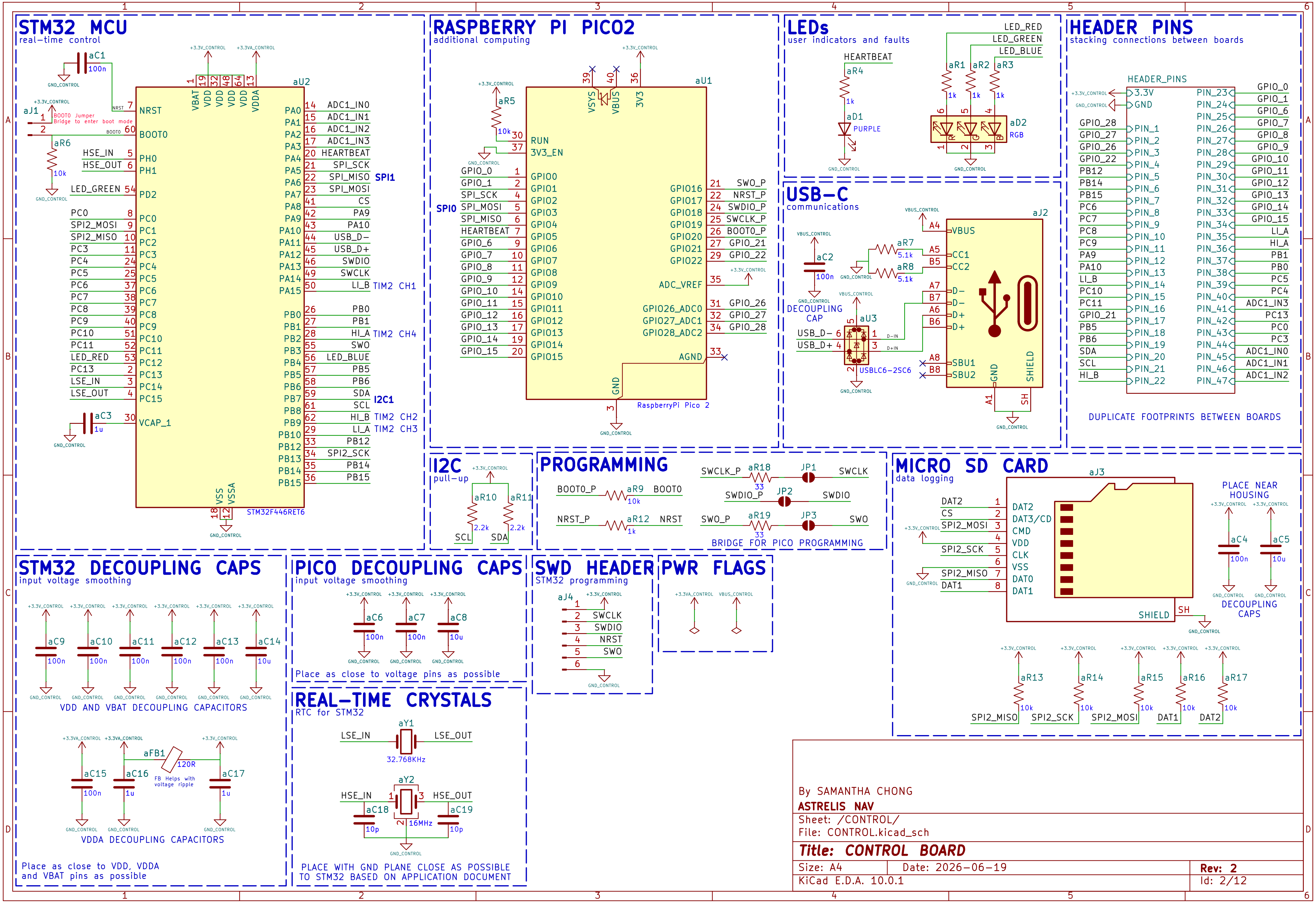


GND DOMAIN ARCHITECTURE
Each subsystem has an isolated GND domain: GND_POWER, GND_CONTROL, GND_SENSOR, GND_TXRX, GND_HEADER. These domains are connected through the stacking connector pins. No single star point.

By SAMANTHA CHONG
ASTRELIS NAV
Sheet: /
File: SWARM_PCB_Rev_2.kicad_sch

Title: SWARM PCB

Size: A4	Date: 2026-06-19	**Rev: 2**
KiCad E.D.A. 10.0.1		Id: 1/12



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ASTRELIS NAV
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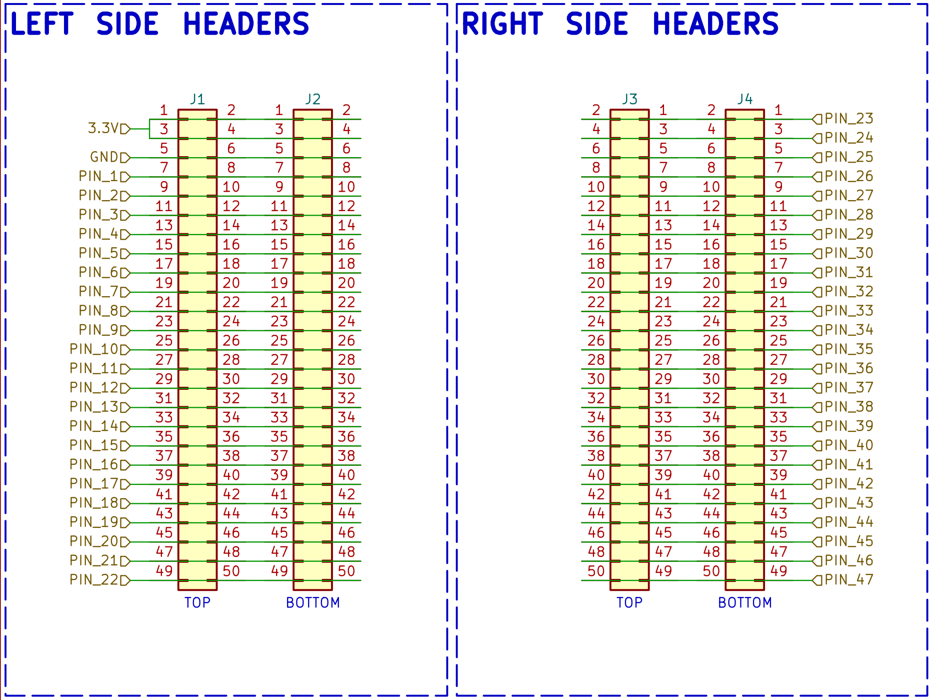
KiCad E.D.A. 10.0.1 | Id: 2/12



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Sheet: /CONTROL/HEADER_PINS/
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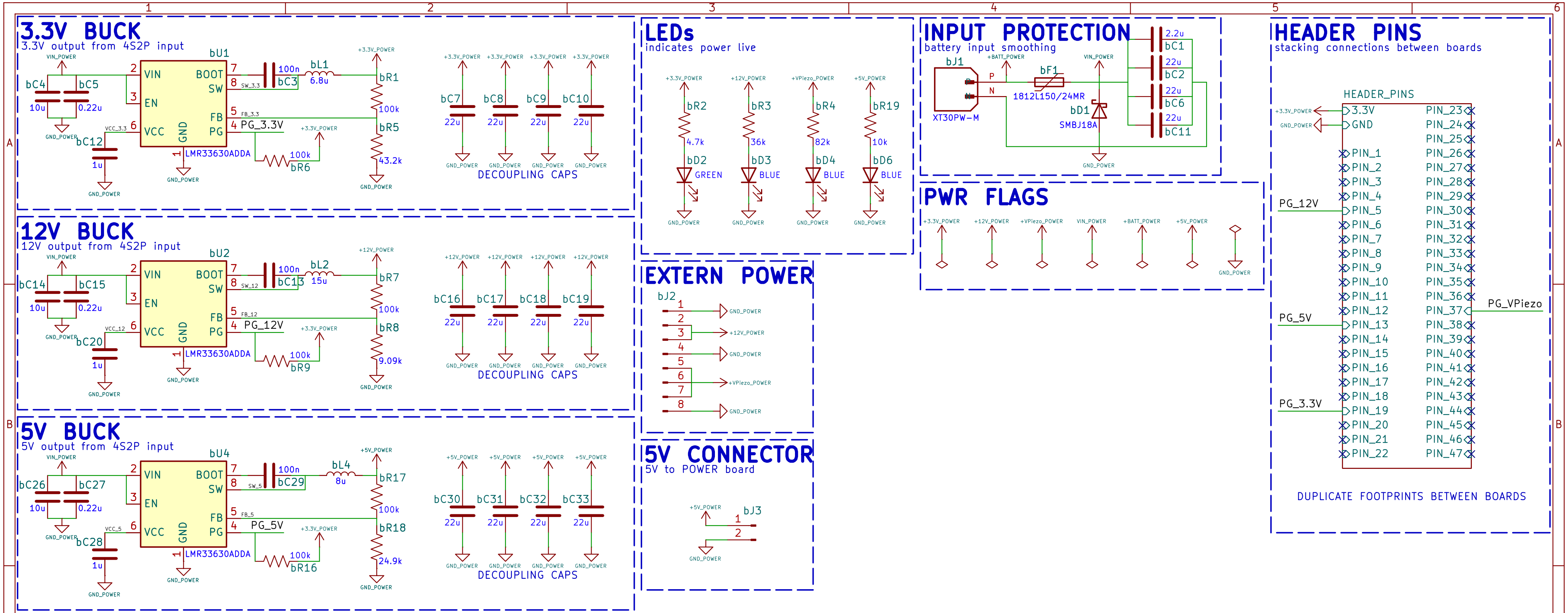
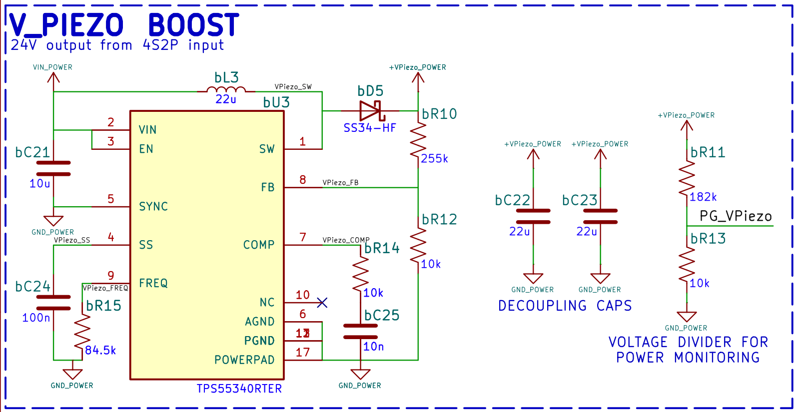
Size: A4 | Date: 2026-06-19 | **Rev:**

KiCad E.D.A. 10.0.1 | Id: 5/12





PG_3.3V → PB6 — digital, open-drain PG pin (pulled up via R6). LOW = 3.3V rail fault.

PG_5V → PA10 — digital, open-drain PG pin (pulled up via R16). LOW = 5V rail fault.

PG_12V → PB12 — digital, open-drain PG pin (pulled up via R9). LOW = 12V rail fault.

PG_VPiezo → PB1 (ADC1_IN9) — analog, R11/R13 voltage divider sampling VPiezo rail.
No digital flag — fault = ADC reading outside expected range.

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Sheet: /POWER/
File: POWER.kicad_sch

Title: POWER BOARD

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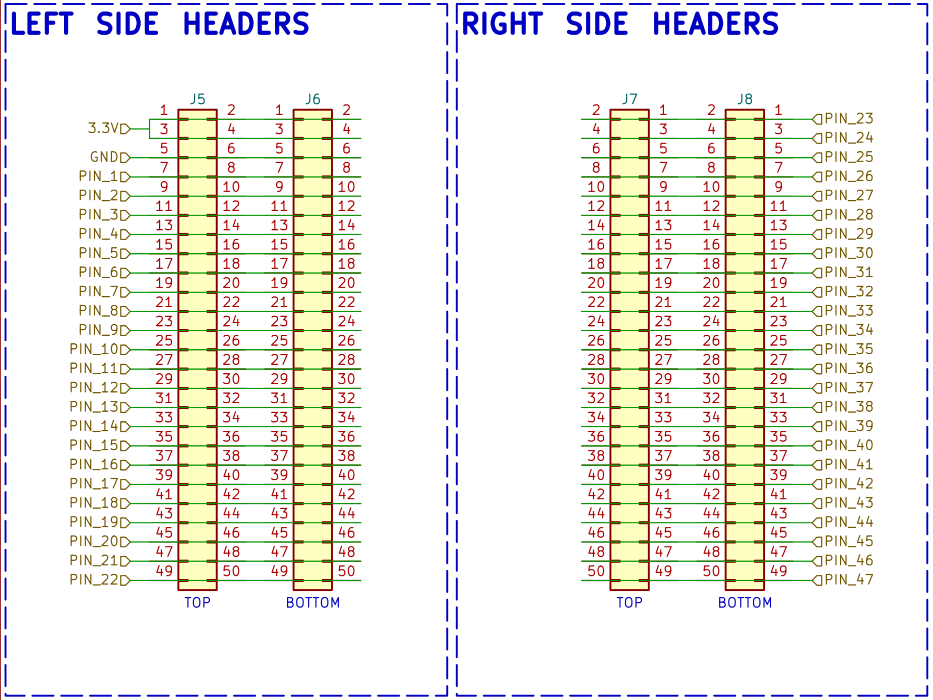
KiCad E.D.A. 10.0.1		Id: 3/12



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ASTRELIS NAV

Sheet: /POWER/HEADER_PINS/

File: HEADER_PINS.kicad_sch

Title:

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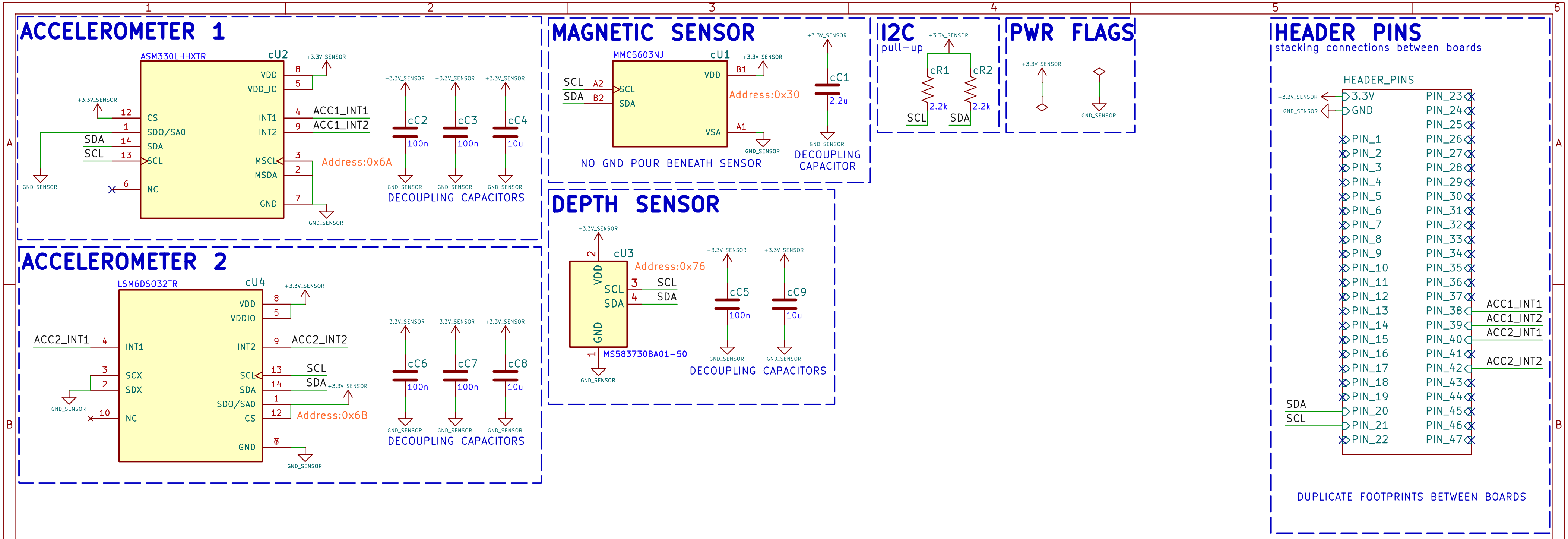
KiCad E.D.A. 10.0.1		Id: 6/12



SCL → PA7 — digital, bit-banged I2C clock (no hardware I2C support on this pin). GPIO open-drain output, external pull-up on bus.

SDA → PA8 — digital, bit-banged I2C data (PA8 only supports I2C3_SCL in hardware, not SDA). GPIO open-drain output, external pull-up on bus.

ACC1_INT1 → PB0 — digital interrupt from Accelerometer 1 (ASM330LHHXTR), INT1. GPIO input + EXTI, edge per sensor config.

ACC1_INT2 → PC5 — digital interrupt from Accelerometer 1 (ASM330LHHXTR), INT2. GPIO input + EXTI.

ACC2_INT1 → PC4 — digital interrupt from Accelerometer 2 (LSM6DSO32TR), INT1. GPIO input + EXTI.

ACC2_INT2 → PC13 — digital interrupt from Accelerometer 2 (LSM6DSO32TR), INT2. GPIO input + EXTI.

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Size: A4 | Date: 2026-06-19 | **Rev:**

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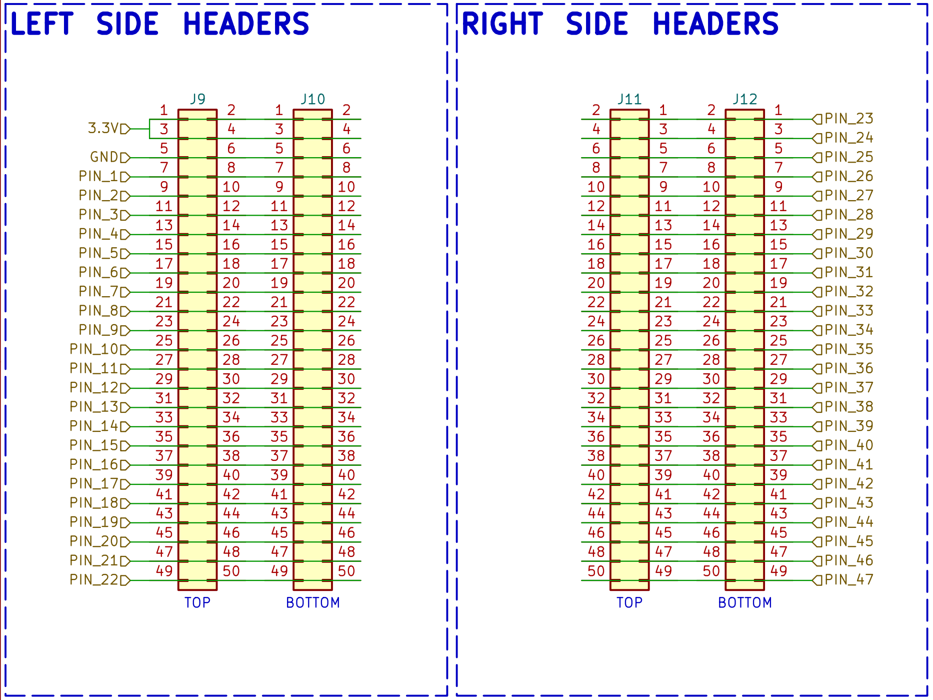

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Sheet: /SENSOR/HEADER_PINS/

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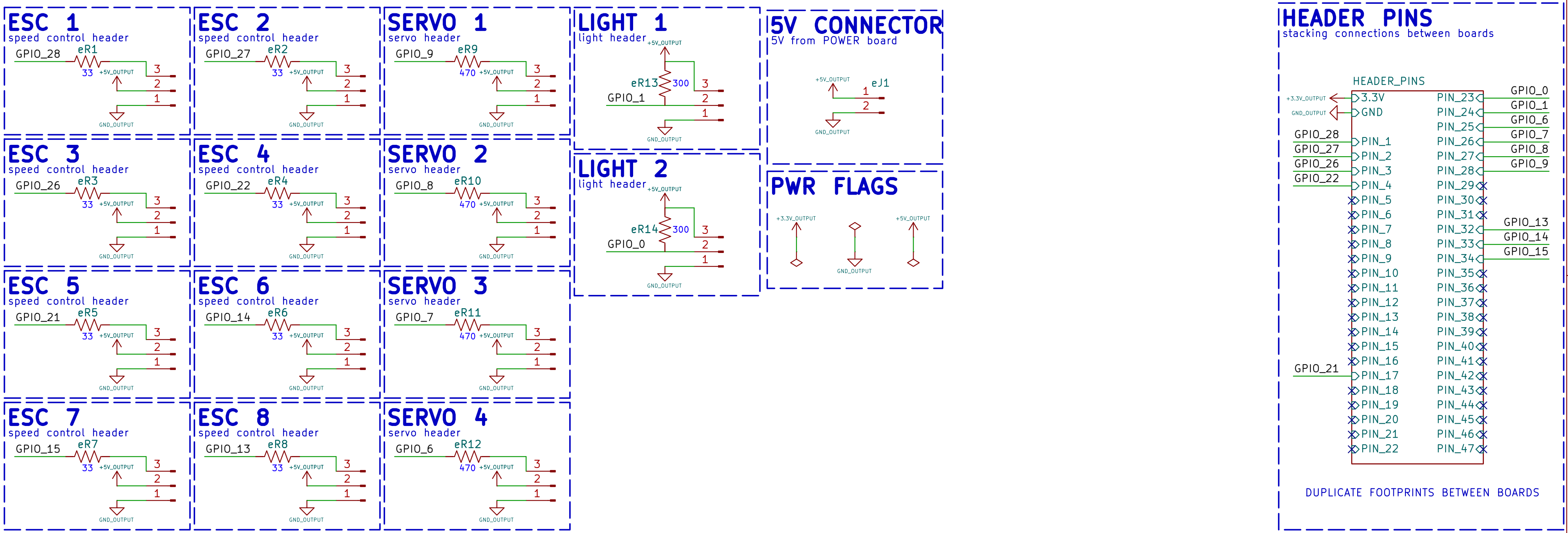

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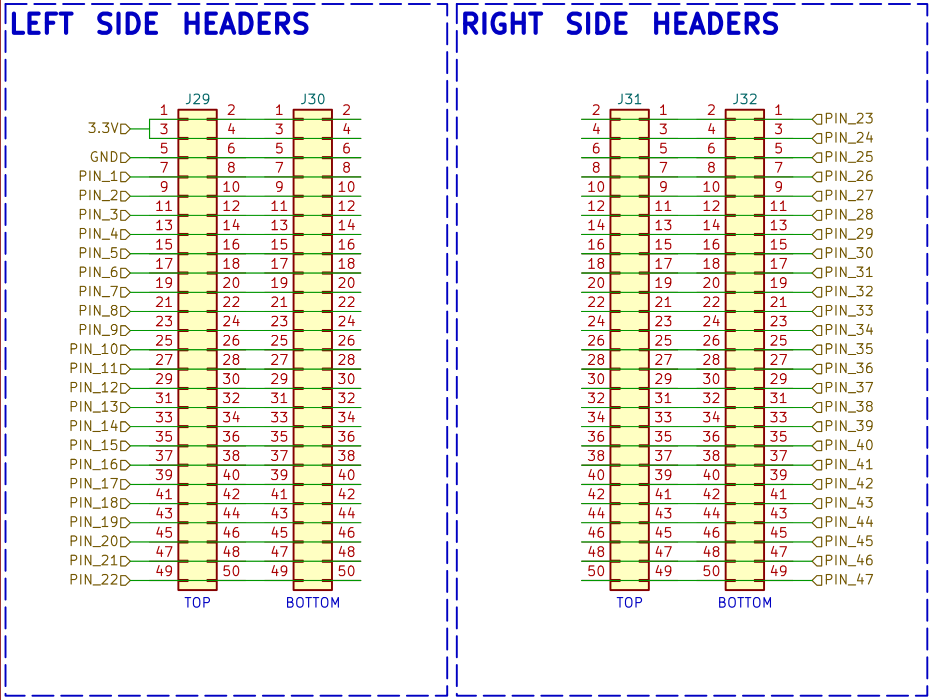
KiCad E.D.A. 10.0.1		Id: 8/12



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ASTRELIS NAV

Sheet: /OUTPUT/HEADER_PINS/
File: HEADER_PINS.kicad_sch

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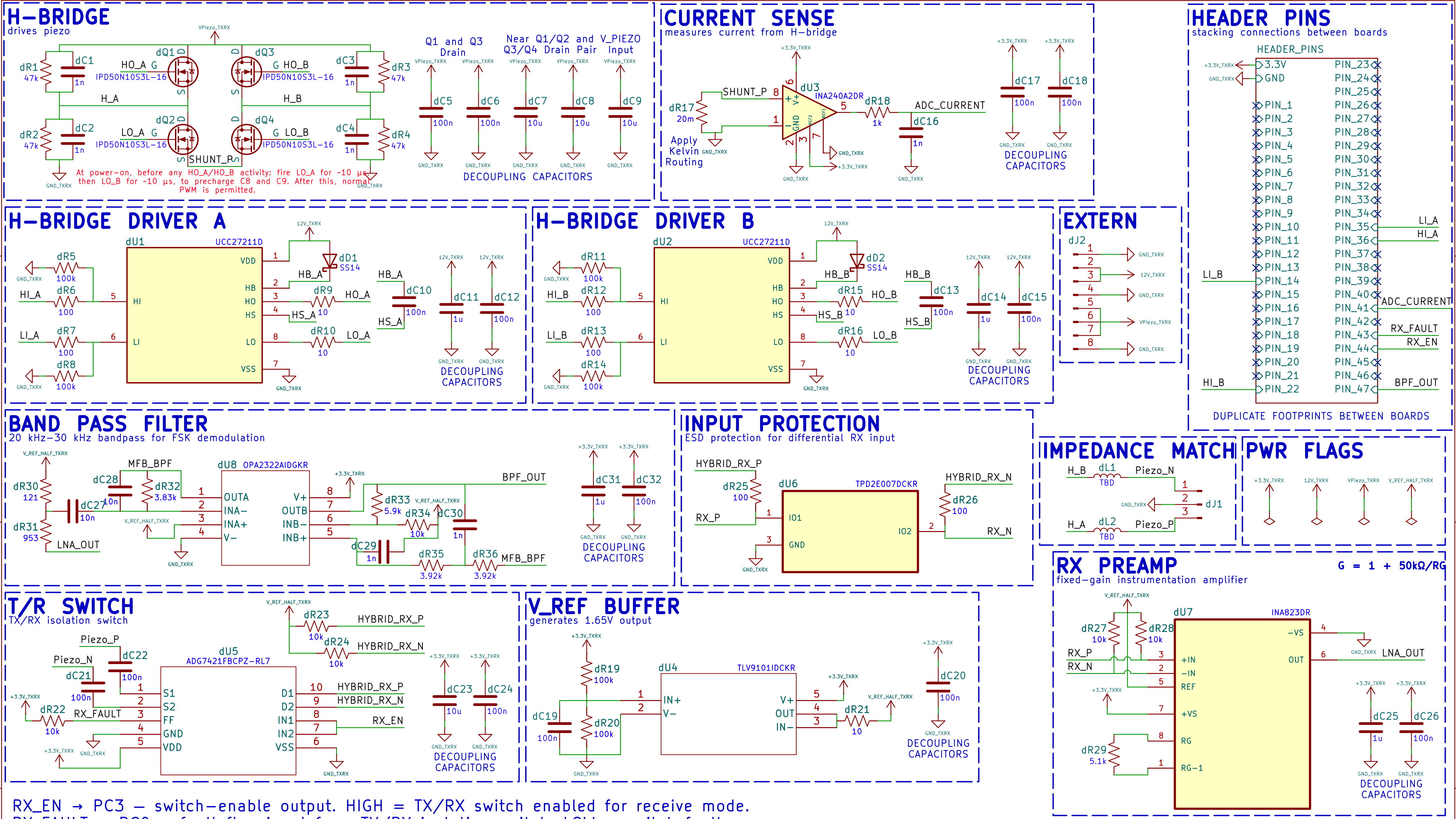
Size: A4 Date: 2026-06-19 **Rev:**
KiCad E.D.A. 10.0.1 Id: 10/12



RX_EN → PC3 — switch-enable output. HIGH = TX/RX switch enabled for receive mode.
RX_FAULT → PC0 — fault flag input from TX/RX isolation switch. LOW = switch fault.

ADC_CURRENT → PA3 (ADC1_IN3) — analog, INA240 current-sense output monitoring H-bridge current.
BPF_OUT → PA2 (ADC1_IN2) — analog, output of 20-30kHz bandpass filter (FSK demod stage).

LI_A → PA5 (TIM2_CH3) — digital PWM output, low-side gate drive to H-bridge driver A.
LI_B → PB0 (TIM2_CH1) — digital PWM output, low-side gate drive to H-bridge driver B.
HI_A → PA13 (TIM2_CH4) — digital PWM output, high-side gate drive to H-bridge driver A.
HI_B → PA6 (TIM2_CH2) — digital PWM output, high-side gate drive to H-bridge driver B.

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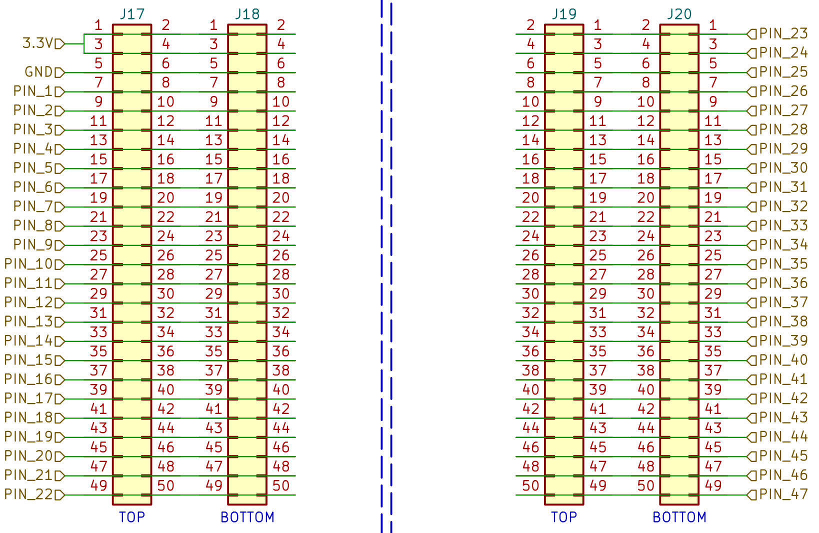
ASTRELIS NAV
Sheet: /TX_RX/
File: TX_RX.kicad_sch

Title: TX & RX

Size: A4	Date: 2026-06-19	**Rev: 2**
KiCad E.D.A. 10.0.1		Id: 9/12

LEFT SIDE HEADERS

RIGHT SIDE HEADERS



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Sheet: /TX_RX/HEADER_PINS/

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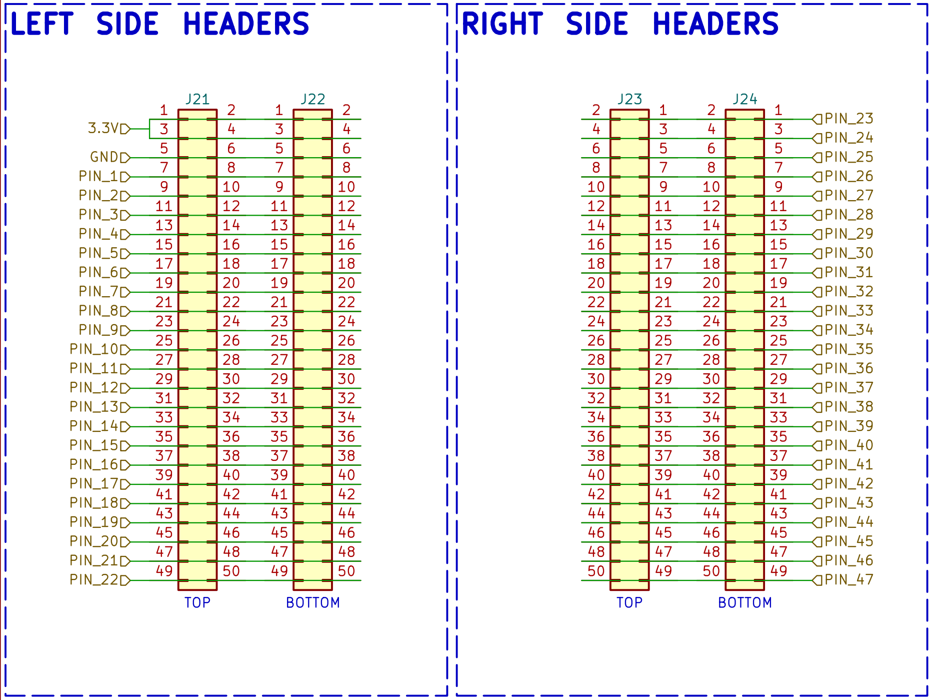

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ASTRELIS NAV

Sheet: /HEADER_PINS/
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KiCad E.D.A. 10.0.1 | Id: 12/12